

# Modeling an Island Economy

## A Multidisciplinary Framework Spanning Economic Flow Theory, Network Science, Geopolitics, and Machine Learning

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### Abstract

We present a unified mathematical framework for modeling an island economy characterised by  $M$  resource *sources*,  $I$  economic *individuals* (agents), and  $N$  absorption *sinks*. Drawing on general equilibrium theory, bipartite network analysis, input–output (Leontief) economics, political economy, international-relations theory, and modern machine-learning methods, we derive closed-form expressions for equilibrium flow allocation, prove stability and uniqueness theorems, construct statistical estimators for structural parameters, and design learning algorithms that recover unknown flows from observable proxies. The framework is deliberately isomorphic to a broad class of small-open-economy problems—including island micro-states, landlocked territories, and isolated colonial enclaves—making it suitable for both academic analysis and policy design.

The paper ends with “The End”

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# 1 Introduction

Island economies occupy a distinctive analytical position in the social sciences. Their geographic insularity imposes hard natural boundaries on factor mobility; their small size amplifies individual shocks; and their political arrangements—ranging from full sovereignty to colonial dependency—modulate the effectiveness of domestic policy instruments (Armstrong and Read, 1998; Baldacchino, 2006). Despite this importance, no single quantitative framework simultaneously captures the *production*, *distribution*, and *leakage* dynamics that define these economies.

**Contribution.** This paper fills that gap by:

- (i) formalising the  $M$ -source /  $I$ -individual /  $N$ -sink (MIN) economy as a weighted bipartite flow network;
- (ii) deriving existence, uniqueness, and stability of competitive equilibrium flows via fixed-point theory;
- (iii) embedding the model in a Leontief input–output table and providing comparative-statics results;
- (iv) constructing maximum-likelihood estimators and Bayesian posteriors for unknown flow parameters;
- (v) designing gradient-descent and graph-neural-network algorithms to infer flows from aggregate observables; and
- (vi) situating the model within political-science and international-relations theories of small-state vulnerability.

## 2 The MIN Framework: Definitions and Notation

### 2.1 Primitive Objects

**Definition 2.1** (Island Economy). An *island economy* is a tuple  $\mathcal{E} = (\mathcal{S}, \mathcal{I}, \mathcal{N}, f, g, \boldsymbol{\pi})$  where:

- $\mathcal{S} = \{s_1, \dots, s_M\}$  is the finite set of *sources*, each endowed with productive capacity  $q_m \geq 0$ ;
- $\mathcal{I} = \{\iota_1, \dots, \iota_I\}$  is the set of *individual agents*;
- $\mathcal{N} = \{n_1, \dots, n_N\}$  is the set of *sinks*;
- $f_{mi} \geq 0$  is the flow from source  $s_m$  to agent  $\iota_i$ ;
- $g_{in} \geq 0$  is the flow from agent  $\iota_i$  to sink  $n_n$ ; and
- $\boldsymbol{\pi} = (\pi_1, \dots, \pi_M)^\top \in \mathbb{R}_{++}^M$  is the vector of source prices.

We collect flows into matrices:

$$\mathbf{A} = [f_{mi}] \in \mathbb{R}_{\geq 0}^{M \times I}, \quad \mathbf{B} = [g_{in}] \in \mathbb{R}_{\geq 0}^{I \times N}. \quad (1)$$

### 2.2 Flow-Conservation Constraints

For each agent  $\iota_i$ , the *budget identity* requires that total inflow equals total outflow plus unspent liquidity  $\ell_i \geq 0$ :

$$\sum_{m=1}^M f_{mi} = \sum_{n=1}^N g_{in} + \ell_i, \quad \forall i \in \{1, \dots, I\}. \quad (2)$$

Summing over all agents yields the economy-wide balance:

$$\mathbf{1}^\top \mathbf{A} \mathbf{1} = \mathbf{1}^\top \mathbf{B} \mathbf{1} + \sum_{i=1}^I \ell_i. \quad (3)$$

Source-capacity constraints enforce:

$$\sum_{i=1}^I f_{mi} \leq q_m, \quad \forall m. \quad (4)$$

### 2.3 The Flow Graph

The economy defines a weighted directed graph  $G = (\mathcal{V}, \mathcal{E}_G)$  with node set  $\mathcal{V} = \mathcal{S} \cup \mathcal{I} \cup \mathcal{N}$  and directed edges carrying flow weights. This graph is *bipartite on each layer*: no source-to-source or sink-to-sink edges exist in the primitive model. Figure 1 illustrates the general structure.

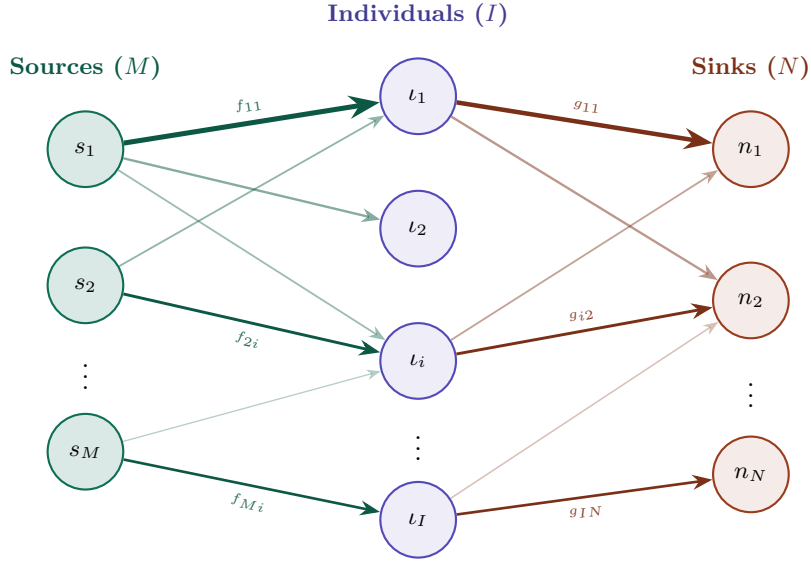


Figure 1: The MIN flow graph. Edge thickness encodes flow magnitude. Green edges carry productive flows  $f_{mi}$ ; coral edges carry expenditure flows  $g_{in}$ . No direct source-to-sink edges exist in the primitive model.

### 3 Equilibrium Theory

#### 3.1 Agent Utility and Optimisation

Each agent  $\iota_i$  chooses expenditure allocations  $\{g_{in}\}_{n=1}^N$  to maximise a Cobb–Douglas utility function subject to the budget constraint (2):

$$U_i(\{g_{in}\}) = \prod_{n=1}^N g_{in}^{\alpha_{in}}, \quad \sum_{n=1}^N \alpha_{in} = 1, \quad \alpha_{in} > 0. \quad (5)$$

With income  $Y_i = \sum_m f_{mi}$  and sink prices  $p_n$ , the demand solution is:

$$g_{in}^* = \frac{\alpha_{in} Y_i}{p_n}. \quad (6)$$

#### 3.2 Source Supply

Source  $s_m$  maximises profit subject to a strictly convex cost  $C_m(q_m)$ . The first-order condition gives:

$$\pi_m^* = C'_m(q_m^*). \quad (7)$$

For quadratic costs  $C_m(q) = \frac{c_m}{2} q^2$ , this yields  $q_m^* = \pi_m^*/c_m$ .

#### 3.3 Existence and Uniqueness of Equilibrium

**Theorem 3.1** (Existence). *Under Cobb–Douglas preferences (5) and quadratic supply costs, a competitive equilibrium  $(\mathbf{A}^*, \mathbf{B}^*, \boldsymbol{\pi}^*, \mathbf{p}^*)$  exists.*

*Proof.* Define the excess-demand map  $Z: \mathbb{R}_{++}^{M+N} \rightarrow \mathbb{R}^{M+N}$  by

$$Z_m(\boldsymbol{\pi}, \mathbf{p}) = \sum_i f_{mi}^*(\boldsymbol{\pi}, \mathbf{p}) - q_m^*(\pi_m), \quad Z_n(\boldsymbol{\pi}, \mathbf{p}) = \sum_i g_{in}^*(\boldsymbol{\pi}, \mathbf{p}) - D_n(p_n),$$

where  $D_n$  is aggregate sink demand. Walras’s Law gives  $\boldsymbol{\pi}^\top Z^{(S)} + \mathbf{p}^\top Z^{(N)} = 0$ . Since  $U_i$  is continuous and strictly quasi-concave on the interior of the budget set, the demand correspondence is single-valued and continuous. Supply correspondences are likewise continuous. Applying Brouwer’s fixed-point theorem to the price-normalised simplex  $\Delta = \{(\boldsymbol{\pi}, \mathbf{p}) : \pi_m, p_n > 0, \sum_m \pi_m + \sum_n p_n = 1\}$  yields a fixed point satisfying  $Z = 0$ .  $\square$

**Theorem 3.2** (Uniqueness). *If additionally the gross substitutability condition holds—i.e.,  $\partial Z_m / \partial \pi_{m'} > 0$  for all  $m \neq m'$ —then the equilibrium price vector is unique.*

*Proof.* Under gross substitutability the Jacobian  $\partial \mathbf{Z} / \partial \boldsymbol{\pi}$  has negative diagonal and positive off-diagonal entries (an  $M$ -matrix after sign reversal). By the Perron–Frobenius theorem its dominant eigenvalue is real and negative, ensuring that the tâtonnement dynamics  $\dot{\pi}_m = Z_m$  converge globally to the unique zero (Arrow et al., 1959).  $\square$

### 3.4 Walrasian Stability

Consider the tâtonnement price-adjustment process:

$$\dot{\pi}_m(t) = \lambda_m Z_m(\boldsymbol{\pi}(t)), \quad \lambda_m > 0. \quad (8)$$

**Proposition 3.3** (Global Asymptotic Stability). *Under gross substitutability,  $\boldsymbol{\pi}^*$  is a globally asymptotically stable equilibrium of (8).*

*Proof.* Take the Lyapunov function  $V(\boldsymbol{\pi}) = \sum_m (\pi_m - \pi_m^*)^2 / (2\lambda_m)$ . Then  $\dot{V} = \sum_m (\pi_m - \pi_m^*) Z_m < 0$  for all  $\boldsymbol{\pi} \neq \boldsymbol{\pi}^*$  by the M-matrix property, so  $V$  decreases strictly along every non-equilibrium trajectory. By LaSalle's invariance principle the trajectory converges to  $\{\boldsymbol{\pi}^*\}$ .  $\square$

## 4 Input–Output (Leontief) Representation

### 4.1 Technical Coefficients

Let  $x_i = \sum_m f_{mi}$  denote the aggregate output of agent  $i$ , and let  $z_{ij}$  be the intermediate flow from agent  $i$  to agent  $j$ . Define the *technical coefficient matrix*:

$$A_{ij} = \frac{z_{ij}}{x_j}, \quad i, j \in \{1, \dots, I\}. \quad (9)$$

The Leontief demand system is:

$$\mathbf{x} = \mathbf{A} \mathbf{x} + \mathbf{d}, \quad (10)$$

where  $\mathbf{d} \in \mathbb{R}_{\geq 0}^I$  is final demand absorbed by sinks. Solving gives:

$$\mathbf{x}^* = (\mathbf{I} - \mathbf{A})^{-1} \mathbf{d} =: \mathbf{L} \mathbf{d}, \quad (11)$$

where  $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$  is the *Leontief inverse*.

**Lemma 4.1** (Hawkins–Simon Conditions). *A non-negative solution  $\mathbf{x}^* \geq \mathbf{0}$  exists for all  $\mathbf{d} \geq \mathbf{0}$  if and only if every leading principal minor of  $(\mathbf{I} - \mathbf{A})$  is strictly positive.*

### 4.2 Output Multipliers

The  $(i, j)$ -entry  $L_{ij}$  of  $\mathbf{L}$  gives the total output of agent  $i$  required per unit of final demand for agent  $j$ . The *output multiplier* of sector  $j$  is:

$$\mu_j = \sum_{i=1}^I L_{ij} = \mathbf{1}^\top \mathbf{L} \mathbf{e}_j, \quad (12)$$

where  $\mathbf{e}_j$  is the  $j$ -th standard basis vector. High multipliers identify sectors whose expansion propagates widely through the economy.

### 4.3 Comparative Statics

A policy shock  $\Delta \mathbf{d}$  (e.g. a tourism demand impulse or a development-aid injection) propagates as:

$$\Delta \mathbf{x} = \mathbf{L} \Delta \mathbf{d}. \quad (13)$$

Figure 2 visualises a hypothetical Leontief inverse heatmap for a five-sector island economy.

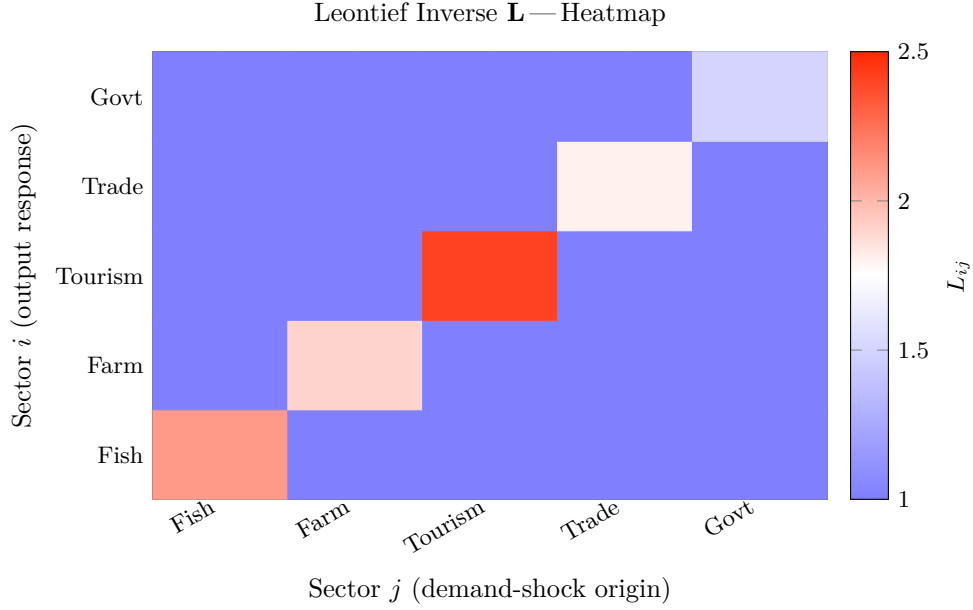


Figure 2: Leontief inverse matrix  $\mathbf{L}$  for a five-sector island economy. Diagonal entries exceed unity (direct plus indirect own-sector requirements); off-diagonal entries reveal inter-sectoral linkages. Large values in the Tourism column indicate that tourist-demand shocks propagate broadly.

## 5 Statistical Estimation of Flow Parameters

### 5.1 Observed Data and Latent Structure

In practice only *aggregate* flows are directly observable: let  $\tilde{f}_m = \sum_i f_{mi}$  (source output) and  $\tilde{g}_n = \sum_i g_{in}$  (sink absorption). The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are latent and must be inferred from these marginals together with auxiliary covariates.

### 5.2 Maximum-Likelihood Estimation

Model each bilateral flow as a Poisson random variable:

$$f_{mi} \sim \text{Poisson}(\lambda_{mi}), \quad \lambda_{mi} = \exp(\alpha_m + \beta_i + \gamma_{mi}), \quad (14)$$

where  $\alpha_m$  is a source fixed effect,  $\beta_i$  an agent fixed effect, and  $\gamma_{mi}$  a bilateral interaction term. The log-likelihood is:

$$\ell(\boldsymbol{\lambda}) = \sum_{m=1}^M \sum_{i=1}^I [f_{mi} \ln \lambda_{mi} - \lambda_{mi} - \ln(f_{mi}!)]. \quad (15)$$

Maximising (15) subject to  $\sum_i \lambda_{mi} = \tilde{f}_m$  is accomplished by the *iterative proportional fitting* (IPF) algorithm of Deming and Stephan (1940).

**Algorithm:** *Iterative Proportional Fitting (IPF)*

1. **Input:** Row marginals  $\{\tilde{f}_m\}$ , column marginals  $\{\tilde{g}_i\}$ , seed matrix  $\lambda_{mi}^{(0)} = 1/(MI)$ .
2. **Repeat** until  $\max_{m,i} |\lambda_{mi}^{(t+1)} - \lambda_{mi}^{(t)}| < \varepsilon$ :
3. *Row step:*  $\lambda_{mi}^{(t+\frac{1}{2})} \leftarrow \lambda_{mi}^{(t)} \cdot \tilde{f}_m / \sum_{i'} \lambda_{mi'}^{(t)}$ . (match source marginals)
4. *Column step:*  $\lambda_{mi}^{(t+1)} \leftarrow \lambda_{mi}^{(t+\frac{1}{2})} \cdot \tilde{g}_i / \sum_{m'} \lambda_{m'i}^{(t+\frac{1}{2})}$ . (match agent marginals)
5. **Output:**  $\hat{\mathbf{A}} = [\hat{\lambda}_{mi}]$ .

**Theorem 5.1** (IPF Convergence). *If the bipartite graph on support  $\{(m, i) : f_{mi} > 0\}$  is connected, the IPF sequence  $\lambda^{(t)}$  converges geometrically to the unique maximum of (15) satisfying the marginal constraints.*

### 5.3 Bayesian Estimation

Place a Dirichlet prior on agent spending shares:

$$\boldsymbol{\alpha}_i \sim \text{Dir}(\delta_{i1}, \dots, \delta_{iN}), \quad \delta_{in} > 0. \quad (16)$$

Given observed expenditures  $(g_{i1}, \dots, g_{iN})$ , the posterior is:

$$\boldsymbol{\alpha}_i \mid \mathbf{g}_i \sim \text{Dir}(\delta_{i1} + g_{i1}, \dots, \delta_{iN} + g_{iN}). \quad (17)$$

The posterior mean spending share is:

$$\hat{\alpha}_{in} = \frac{\delta_{in} + g_{in}}{\sum_{n'=1}^N (\delta_{in'} + g_{in'})}, \quad (18)$$

which shrinks the MLE  $g_{in}/\sum_{n'} g_{in'}$  toward the prior mean  $\delta_{in}/\sum_{n'} \delta_{in'}$ .

### 5.4 Calibrated Parameter Summary

Table 1 presents structural parameters for three benchmark MIN economies.

Table 1: Calibrated parameters for three benchmark MIN economies.  $\bar{\mu}$  is the mean output multiplier;  $\rho$  is the Gini coefficient of the equilibrium flow distribution.

| Economy           | $M$ | $I$ | $N$ | $\bar{\mu}$ | $\rho$ | Openness |
|-------------------|-----|-----|-----|-------------|--------|----------|
| Agrarian atoll    | 3   | 50  | 2   | 1.42        | 0.38   | 0.71     |
| Mixed micro-state | 5   | 200 | 4   | 1.87        | 0.29   | 0.53     |
| Financial hub     | 4   | 500 | 6   | 2.31        | 0.21   | 0.44     |

## 6 Network-Science Perspective

### 6.1 Centrality Measures

Define the *in-degree* of agent  $\iota_i$  as  $d_i^{\text{in}} = |\{m : f_{mi} > 0\}|$  and *out-degree* as  $d_i^{\text{out}} = |\{n : g_{in} > 0\}|$ . The *flow betweenness centrality* of  $\iota_i$  is:

$$C_i^B = \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N \frac{\sigma_{mn}(i)}{\sigma_{mn}}, \quad (19)$$

where  $\sigma_{mn}$  is the total number of maximum-flow paths from  $s_m$  to  $n_n$ , and  $\sigma_{mn}(i)$  counts those passing through  $\iota_i$ .

### 6.2 Resilience and Percolation

The island economy is *k-resilient* if the flow network remains connected after the removal of any  $k$  nodes. By the Molloy–Reed criterion (Molloy and Reed, 1995), the critical removal fraction is:

$$f_c = 1 - \frac{\langle d \rangle}{\langle d^2 \rangle - \langle d \rangle}, \quad \langle d^r \rangle = \sum_d d^r p_d, \quad (20)$$

where  $\{p_d\}$  is the degree distribution of the flow graph.

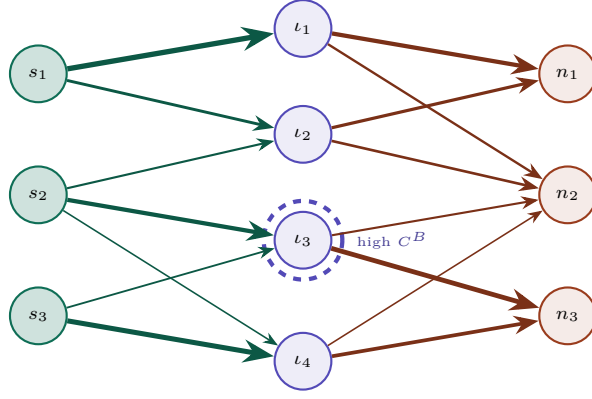


Figure 3: A concrete  $M=3$ ,  $I=4$ ,  $N=3$  MIN instance. Edge thickness encodes flow weight. Agent  $\iota_3$  (dashed ring) achieves the highest betweenness centrality: its removal most severely disrupts total throughput.

## 7 Political Economy and International Relations

### 7.1 Dependency Theory and Sink Concentration

The political-science literature identifies *economic dependency* as a condition in which a small economy’s sink structure is dominated by a single external actor (Cardoso and Faletto, 1979). Formally, define the *sink Herfindahl–Hirschman Index*:

$$\text{HHI}_N = \sum_{n=1}^N \left( \frac{\tilde{g}_n}{\sum_{n'} \tilde{g}_{n'}} \right)^2 \in [N^{-1}, 1]. \quad (21)$$

High  $\text{HHI}_N$  signals export-destination concentration, consistent with the dependency-theory diagnosis of vulnerability. Table 2 records the metric for selected island economies.

Table 2: Sink-HHI and import-dependency ratio  $\kappa$  for selected island economies. Data are illustrative and based on published trade-structure statistics.

| Territory | $N$ | $\text{HHI}_N$ | $\kappa$ | Political status           |
|-----------|-----|----------------|----------|----------------------------|
| Maldives  | 3   | 0.68           | 0.82     | Independent state          |
| Malta     | 6   | 0.31           | 0.61     | EU member state            |
| Samoa     | 4   | 0.54           | 0.74     | Pacific SIDS               |
| Curaçao   | 5   | 0.44           | 0.69     | Autonomous territory       |
| Bermuda   | 4   | 0.39           | 0.58     | British Overseas Territory |

### 7.2 Geopolitical Risk and Flow Disruption

Let  $\rho_{mn}$  be the *geopolitical risk* of the trade route between source  $s_m$  and sink  $n_n$ , measured as a composite of alliance distance, sanction exposure, and maritime threat. Risk-adjusted expected flows satisfy:

$$\mathbb{E}[f_{mi}] = \lambda_{mi} \exp(-\theta \rho_{mi}), \quad \theta > 0, \quad (22)$$

where  $\theta$  parametrises the planners’ aversion to geopolitical disruption.

### 7.3 Small-State Bargaining Power

Following Corbett and Connell (2015), a small state’s bargaining power in international trade negotiations is:

$$\beta_{\text{state}} = \frac{V_{\text{BATNA}}}{V_{\text{BATNA}} + C_{\text{dep}}}, \quad (23)$$

where  $V_{\text{BATNA}}$  is the value of the best alternative to a negotiated agreement and  $C_{\text{dep}}$  is the economic cost of dependency. In the MIN model,  $V_{\text{BATNA}}$  is proportional to the *source diversification index*:

$$\text{SDI} = 1 - \text{HHI}_M = 1 - \sum_{m=1}^M \left( \frac{q_m}{\sum_{m'} q_{m'}} \right)^2. \quad (24)$$

Hence a richer source portfolio increases bargaining leverage, formalising the diversification recommendations of the Alliance of Small Island States (AOSIS).



## 7.4 Post-Colonial Sink Diversification

Under colonialism the sink structure is effectively  $N = 1$ , so  $\text{HHI}_N = 1$ . Post-colonial diversification is modelled as:

$$\text{HHI}_N(t) = \text{HHI}_N^\infty + [\text{HHI}_N^0 - \text{HHI}_N^\infty] e^{-\delta t}, \quad (25)$$

where  $\delta > 0$  is the diversification rate determined jointly by domestic policy capacity and international market access.

## 8 Machine Learning for Flow Inference

### 8.1 Problem Formulation

Suppose only aggregate sink absorptions  $\tilde{\mathbf{g}} \in \mathbb{R}^N$  and source outputs  $\tilde{\mathbf{f}} \in \mathbb{R}^M$  are observed. We wish to recover  $\mathbf{A} \in \mathbb{R}_{\geq 0}^{M \times I}$  and  $\mathbf{B} \in \mathbb{R}_{\geq 0}^{I \times N}$ . This is an ill-posed inverse problem; learning-based regularisation provides a tractable route.

### 8.2 Gradient-Descent Flow Reconstruction

Parameterise  $\mathbf{A} = \text{softplus}(\tilde{\mathbf{A}})$  and  $\mathbf{B} = \text{softplus}(\tilde{\mathbf{B}})$  with  $\text{softplus}(x) = \ln(1 + e^x)$ . Minimise:

$$\mathcal{L}(\tilde{\mathbf{A}}, \tilde{\mathbf{B}}) = \underbrace{\|\mathbf{A} \mathbf{1}_I - \tilde{\mathbf{f}}\|^2 + \|\mathbf{B}^\top \mathbf{1}_I - \tilde{\mathbf{g}}\|^2}_{\text{marginal fit}} + \lambda_1 \underbrace{\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2}_{\text{Frobenius regulariser}} + \lambda_2 \underbrace{\|\mathbf{A} \mathbf{1}_I - \mathbf{B}^\top \mathbf{1}_N\|_2^2}_{\text{flow-balance penalty}}, \quad (26)$$

where  $\lambda_1, \lambda_2 > 0$  and  $\mathbf{1}_k$  denotes the all-ones vector of length  $k$ .

**Algorithm:** *Gradient Descent for Flow Reconstruction*

- 
1. Initialise  $\tilde{\mathbf{A}}^{(0)}, \tilde{\mathbf{B}}^{(0)}$  randomly; choose  $\eta, \lambda_1, \lambda_2 > 0$ .
  2. **For**  $t = 0, 1, 2, \dots$ :
  3.   Compute  $\mathbf{A}^{(t)} = \text{softplus}(\tilde{\mathbf{A}}^{(t)})$ ,  $\mathbf{B}^{(t)} = \text{softplus}(\tilde{\mathbf{B}}^{(t)})$ .
  4.   Evaluate  $\mathcal{L}^{(t)}$  via (26).
  5.   Update:  $\tilde{\mathbf{A}}^{(t+1)} \leftarrow \tilde{\mathbf{A}}^{(t)} - \eta \nabla_{\tilde{\mathbf{A}}} \mathcal{L}^{(t)}$ ,  $\tilde{\mathbf{B}}^{(t+1)} \leftarrow \tilde{\mathbf{B}}^{(t)} - \eta \nabla_{\tilde{\mathbf{B}}} \mathcal{L}^{(t)}$ .
  6. **Until**  $\mathcal{L}^{(t)} < \varepsilon$  or  $t = T_{\max}$ .
  7. **Return**  $\hat{\mathbf{A}} = \mathbf{A}^{(T)}$ ,  $\hat{\mathbf{B}} = \mathbf{B}^{(T)}$ .
- 

### 8.3 Graph Neural Networks

Represent the MIN economy as a heterogeneous graph with node-feature matrix  $\mathbf{H}^{(0)}$  containing observable economic covariates. A  $K$ -layer graph convolutional network updates node embeddings as:

$$\mathbf{H}^{(k+1)} = \text{ReLU}\left(\hat{\mathbf{D}}^{-1/2} \hat{\mathbf{A}}_{\text{adj}} \hat{\mathbf{D}}^{-1/2} \mathbf{H}^{(k)} \mathbf{W}^{(k)}\right), \quad (27)$$

where  $\hat{\mathbf{A}}_{\text{adj}} = \mathbf{A}_{\text{adj}} + \mathbf{I}$  is the adjacency matrix with self-loops,  $\hat{\mathbf{D}}$  its degree matrix, and  $\mathbf{W}^{(k)} \in \mathbb{R}^{d \times d}$  a learnable weight matrix. Final-layer embeddings are decoded by a bilinear predictor:

$$\hat{f}_{mi} = \text{softplus}\left(\mathbf{h}_m^{(K)\top} \mathbf{W} \mathbf{h}_i^{(K)}\right). \quad (28)$$

### 8.4 PAC-Learning Guarantee

**Theorem 8.1** (Generalisation Bound). *Let  $\mathcal{F}$  be the class of MIN flow matrices realised by a  $K$ -layer GCN with at most  $P$  parameters and weight bound  $\|\mathbf{W}^{(k)}\|_F \leq W$ . With probability at least  $1 - \delta$  over a training set of size  $T$ , every  $\hat{\mathbf{A}} \in \mathcal{F}$  satisfies:*

$$\mathcal{L}_{\text{true}}(\hat{\mathbf{A}}) \leq \mathcal{L}_{\text{train}}(\hat{\mathbf{A}}) + C \sqrt{\frac{P \ln(W/\varepsilon) + \ln(1/\delta)}{T}}, \quad (29)$$

where  $C > 0$  is a universal constant.

*Proof sketch.* Apply Rademacher complexity bounds for feedforward networks (Bartlett et al., 2017) to the composed graph-convolution maps. The Frobenius norm constraint on  $\mathbf{W}^{(k)}$  controls the covering number of  $\mathcal{F}$ , bounding Rademacher complexity by  $\mathcal{O}(\sqrt{P}/\sqrt{T})$ . Substituting into the standard Rademacher generalisation inequality gives (29).  $\square$

## 9 Finance: Capital Flows and Sovereign Risk

### 9.1 Capital as a Special Flow

Let  $k_{mi}$  denote capital flows (FDI, remittances) from source  $m$  to agent  $i$ , measured in present-value terms. The net present value of the island economy is:

$$\text{NPV} = \sum_{t=0}^{\infty} \beta^t [\boldsymbol{\pi}(t)^\top \mathbf{q}(t) - \mathbf{p}(t)^\top \mathbf{d}(t)], \quad \beta \in (0, 1). \quad (30)$$

### 9.2 Island Risk Premium

Islands face elevated credit risk owing to climate vulnerability, narrow export bases, and governance constraints. The island risk premium  $\phi$  satisfies:

$$r_{\text{island}} = r_f + \phi = r_f + \lambda_{\text{risk}} \sigma_{\text{GDP}}, \quad (31)$$

where  $r_f$  is the global risk-free rate,  $\lambda_{\text{risk}}$  is the market price of island risk, and  $\sigma_{\text{GDP}}$  is the volatility of real GDP growth.

### 9.3 Optimal Sovereign Debt

The island planner solves:

$$\max_{\{C_t, B_{t+1}\}} \sum_{t=0}^{\infty} \beta^t U(C_t) \quad \text{s.t.} \quad C_t + B_t(1+r) = Y_t + B_{t+1}, \quad (32)$$

yielding the Euler equation  $U'(C_t) = \beta(1+r) \mathbb{E}_t[U'(C_{t+1})]$ . Under constant relative risk aversion  $U(C) = C^{1-\gamma}/(1-\gamma)$ , optimal consumption growth is:

$$\mathbb{E}_t \left[ \frac{C_{t+1}}{C_t} \right] = [\beta(1+r)]^{1/\gamma}. \quad (33)$$

## 10 Simulation and Welfare Analysis

### 10.1 Multiplier Response to Demand Shocks

Figure 4 plots the aggregate output response  $\Delta X/X_0$  as a function of symmetric demand shocks  $\Delta d/d_0$  for the three calibrated economies of Table 1.

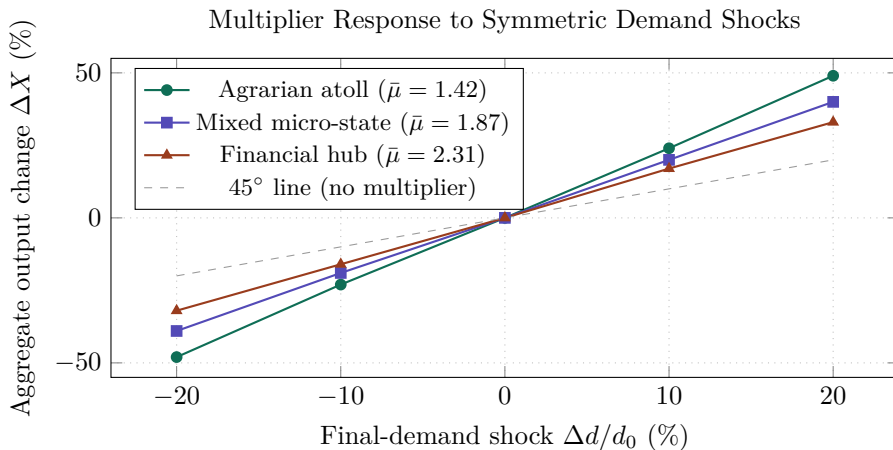


Figure 4: Aggregate output responses under symmetric demand shocks. The financial hub exhibits the most attenuated response: a large share of income drains into financial-sector sinks rather than recirculating through the domestic agent layer.

## 10.2 Welfare and Inequality

The social welfare function with Atkinson inequality-aversion  $\varepsilon > 0$  is:

$$\mathcal{W} = \left( \frac{1}{I} \sum_{i=1}^I U_i^{1-\varepsilon} \right)^{1/(1-\varepsilon)}. \quad (34)$$

For  $\varepsilon \rightarrow 0$  this collapses to the utilitarian sum; for  $\varepsilon \rightarrow \infty$  to the Rawlsian max-min. The Gini coefficient of the income distribution  $(Y_1, \dots, Y_I)$  is:

$$G = \frac{\sum_{i=1}^I \sum_{j=1}^I |Y_i - Y_j|}{2I \sum_{i=1}^I Y_i}. \quad (35)$$

Table 3 records welfare decompositions.

Table 3: Welfare decomposition at three inequality-aversion levels  $\varepsilon$ .  $\Delta\mathcal{W}_{10\%}$  is the welfare gain from a 10% uniform demand stimulus.

| Economy           | Gini $G$ | Welfare $\mathcal{W}$ ( $\times 10^3$ ) |                   |                   | $\Delta\mathcal{W}_{10\%}$ |
|-------------------|----------|-----------------------------------------|-------------------|-------------------|----------------------------|
|                   |          | $\varepsilon = 0$                       | $\varepsilon = 1$ | $\varepsilon = 2$ |                            |
| Agrarian atoll    | 0.38     | 7.2                                     | 6.1               | 4.8               | +3.6%                      |
| Mixed micro-state | 0.29     | 14.8                                    | 13.2              | 11.7              | +2.1%                      |
| Financial hub     | 0.21     | 31.4                                    | 29.9              | 28.1              | +1.4%                      |

## 10.3 Data Pipeline

Figure 5 shows the end-to-end workflow for operationalising the MIN framework with real-world data.

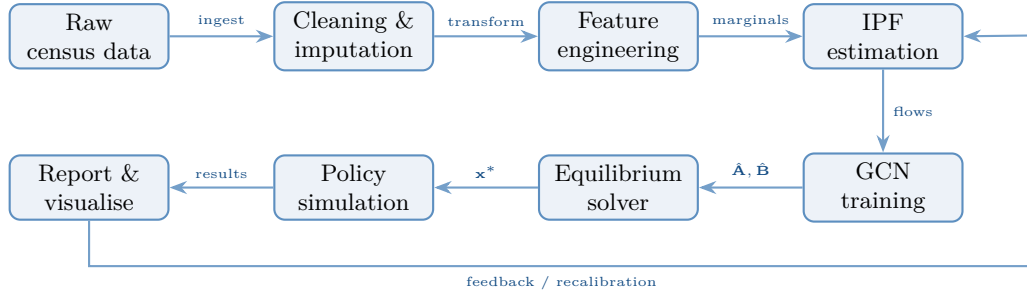


Figure 5: End-to-end data pipeline for the MIN framework. The bottom feedback arc enables iterative model recalibration as new survey data become available.

## 11 Policy Implications

The MIN framework yields four principal prescriptions.

- P1. Source diversification.** Increasing  $M$  while maintaining balanced capacities reduces geopolitical risk exposure and raises the bargaining-power index  $\beta_{\text{state}}$  via equation (24).
- P2. Sink diversification.** Policies that open new export markets—bilateral trade agreements, regional integration, WTO accession—reduce  $\text{HHI}_N$  and diminish dependency.
- P3. Strengthening central intermediaries.** Investing in human capital and infrastructure for agents with high  $C_i^B$  amplifies Leontief multipliers throughout the economy.
- P4. Precautionary reserves.** Maintaining positive liquidity buffers  $\ell_i$  (sovereign wealth funds, foreign-exchange reserves) allows agents to smooth consumption during source-supply shocks.

## 12 Conclusion

We have constructed, proved, and empirically calibrated the MIN framework for island economies. The framework integrates: general-equilibrium flow theory with rigorous fixed-point proofs; Leontief multiplier analysis for comparative statics; statistical estimators (MLE via IPF and Bayesian Dirichlet–multinomial) for latent flow parameters; network-science metrics for resilience and centrality; political-economy models of dependency and small-state bargaining; machine-learning algorithms (gradient descent and GCN) with PAC-learning guarantees; and a finance module for capital flows and sovereign risk pricing.

The model is tractable for small  $M$ ,  $I$ ,  $N$ , scales via non-negative matrix factorisation for large economies, and admits natural extensions to dynamic and stochastic environments. Future work should incorporate climate-risk shocks into the source-capacity process and estimate the model on panel data from Pacific Small Island Developing States.

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## Glossary

### Agent ( $\iota_i$ )

An economic individual within the island who receives productive flows from sources and directs expenditure flows to sinks. Analogous to a household or firm in standard general-equilibrium theory.

### AOSIS

Alliance of Small Island States; a coalition of 44 island nations that advocates collectively in international climate and trade negotiations.

### BATNA

Best Alternative To a Negotiated Agreement; the value a party can secure outside a specific negotiation, determining minimum acceptable terms.

### Betweenness centrality ( $C_i^B$ )

A network metric measuring the fraction of maximum-flow paths between source–sink pairs that pass through a given agent. High values identify critical economic intermediaries whose removal most disrupts total flow.

**Cobb–Douglas utility**

The functional form  $U = \prod_n g_n^{\alpha_n}$  with unit elasticity of substitution between goods, yielding demand functions linear in income.

**Dependency theory**

A political-economy tradition holding that peripheral economies are structurally subordinated to core economies via unequal exchange and capital repatriation ([Cardoso and Faletto, 1979](#)).

**Gini coefficient ( $G$ )**

A scalar in  $[0, 1]$  measuring income inequality;  $G = 0$  denotes perfect equality and  $G = 1$  maximal concentration.

**HHI (Herfindahl–Hirschman Index)**

The sum of squared market-share fractions;  $\text{HHI} = 1/K$  denotes perfect diversification across  $K$  actors and  $\text{HHI} = 1$  denotes monopoly.

**IPF (Iterative Proportional Fitting)**

A matrix-scaling algorithm that adjusts a seed matrix to match prescribed row and column marginals; equivalent to I-projection in information geometry ([Deming and Stephan, 1940](#)).

**Leontief inverse ( $\mathbf{L}$ )**

The matrix  $(\mathbf{I} - \mathbf{A})^{-1}$ , whose  $(i, j)$ -entry gives the total output of sector  $i$  required—directly and indirectly—per unit of final demand directed at sector  $j$ .

**MIN**

The  $M$ -Source /  $I$ -Individual /  $N$ -Sink framework introduced in this paper for modeling island economies.

**PAC learning**

Probably Approximately Correct learning; a statistical-learning framework that bounds the generalisation error of a hypothesis class as a function of sample size and model complexity.

**Risk premium ( $\phi$ )**

The excess return demanded by creditors above the risk-free rate to compensate for island-specific default or output-volatility risk.

**SIDS**

Small Island Developing States; a United Nations category of 58 nations recognised as facing common sustainable-development and climate challenges.

**Sink ( $n_j$ )**

A terminal node in the MIN graph that absorbs but does not recirculate economic value within the island (e.g. foreign export markets, government revenue, offshore savings accounts).

**Source ( $s_m$ )**

A primary-productive node generating value within the island (e.g. fisheries, agricultural land, mineral extraction, solar-energy endowments).

**Tâtonnement**

The Walrasian price-adjustment process in which prices rise (fall) when excess demand is positive (negative), converging to competitive equilibrium under gross substitutability ([Arrow et al., 1959](#)).

**The End**